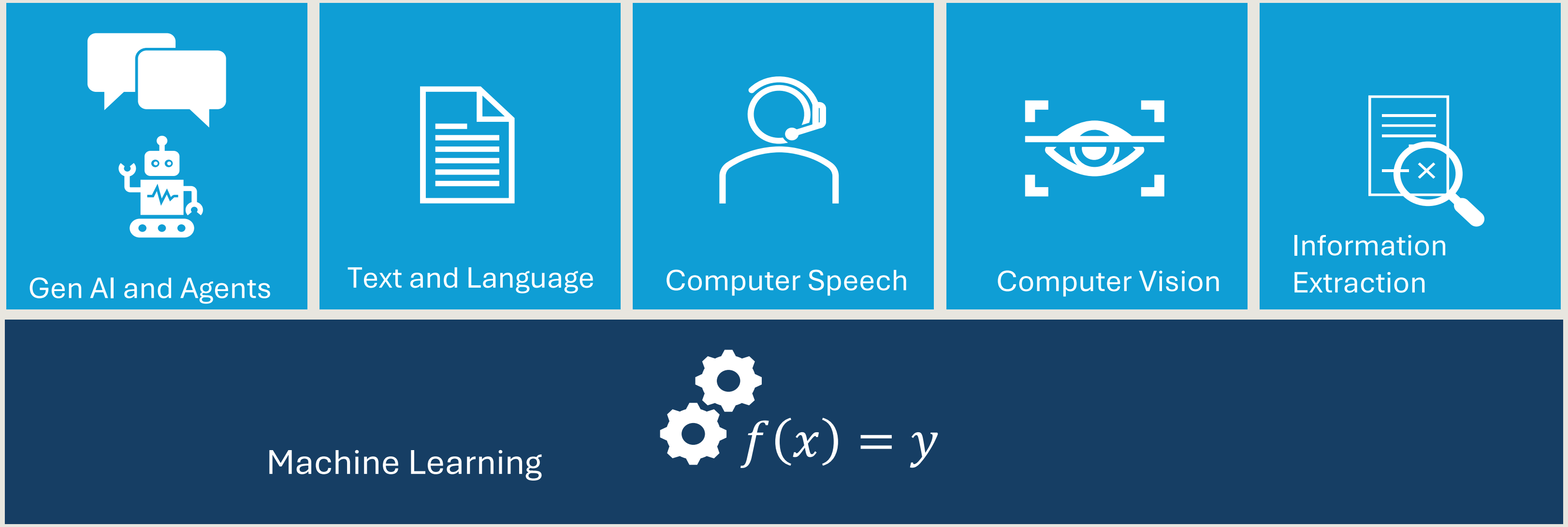
Microsoft

Develop natural language solutions in Azure

https://aka.ms/mslearn-language

Agenda

1. Develop generative AI apps in Azure
2. Develop AI agents in Azure
3. Develop natural language solutions in Azure
4. Extract insights from visual data on Azure



Our focus today

Generative AI and Agents

Text and Language

Computer Speech

Computer Vision

Information Extraction

$f(x) = y$

Machine Learning

van Moof

Analyze text with Azure Language in Foundry Tools

https://aka.ms/mslearn-analyze-text

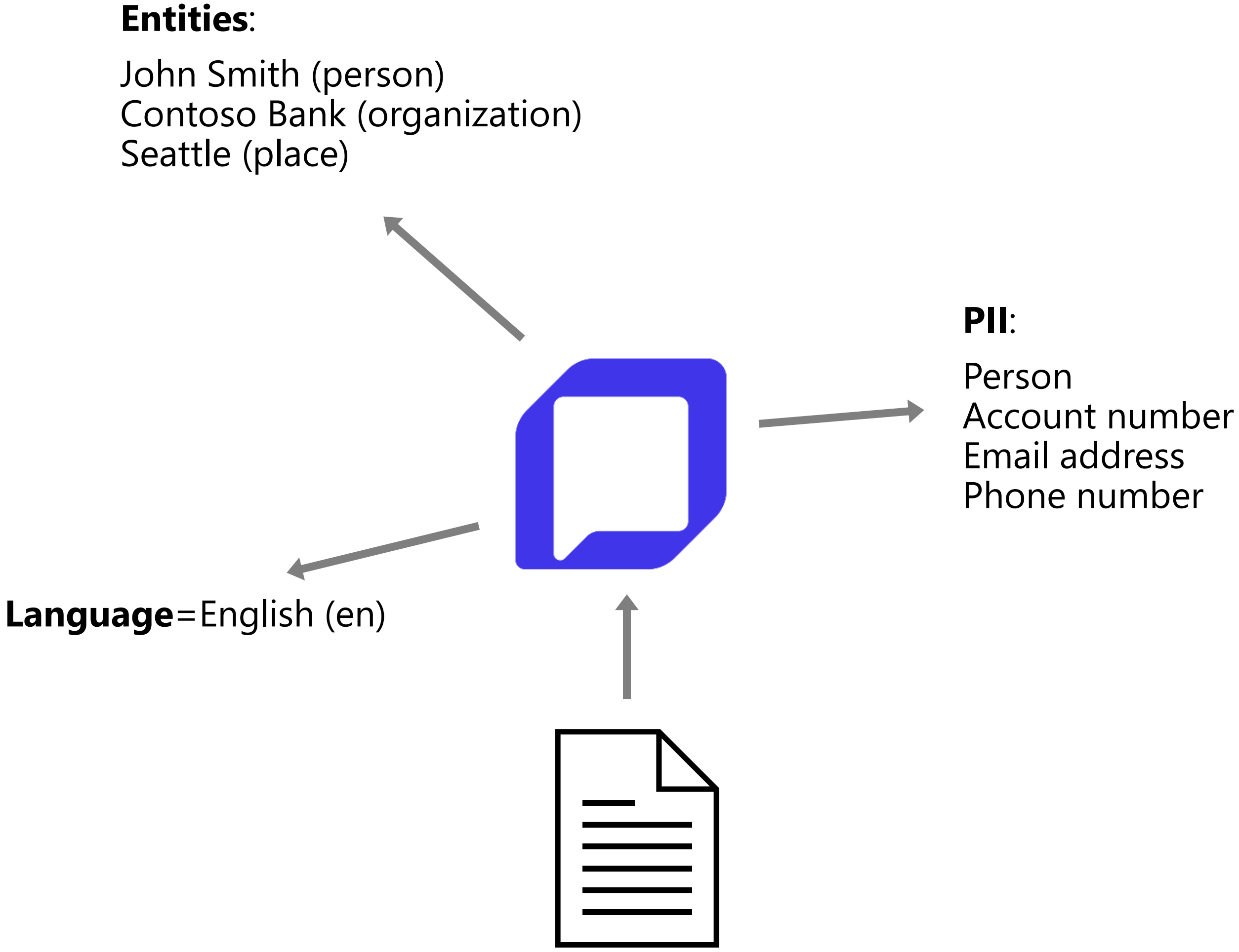
Esperanto

Azure Language in Foundry Tools

NLP

Prebuilt models for:

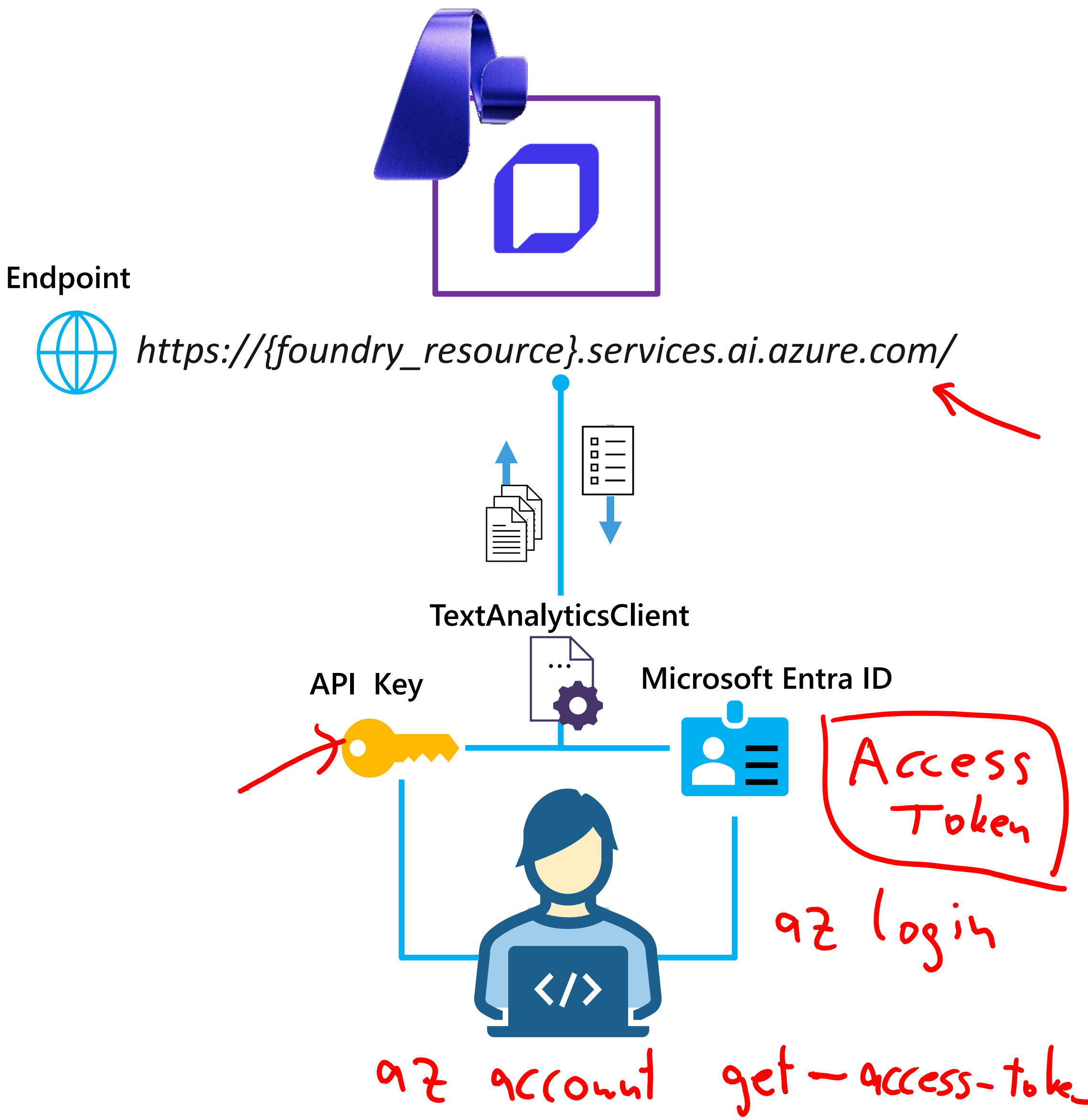
- Language detection
- Named entity recognition NER
- PII detection
- Sentiment Analysis



Entra ID

Using the Text Analytics API

1. Connect to the API
   - **Endpoint:**
     - Microsoft Foundry resource*
   - **Credentials:**
     - Microsoft Foundry project API key
     - Microsoft Entra ID
2. Use a *TextAnalyticsClient* to call the appropriate function for your task
   - **Request**: One or more text documents
   - **Response**: Task-specific collection of results (one per document)



* cognitiveservices.azure.com is also supported

Exercise

Analyze text

In this exercise, you'll use Azure Language to:

- Detect Language
- Extract Entities
- Extract PII

Start the exercise at:

13. Analyze text

1 h

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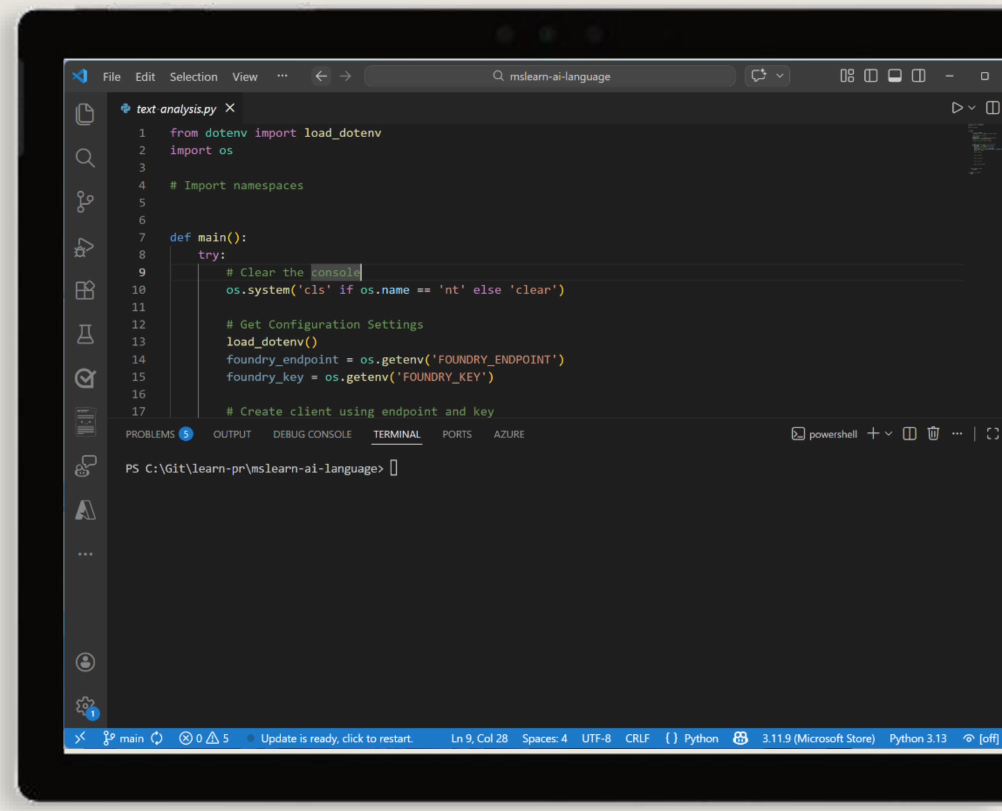


Knowledge check

1 **How should you create an application that analyzes news articles and extracts key people, places, and dates that are mentioned for indexing?**

☐ Use a generative AI model with a custom function tool that matches strings using a regular expression.

☐ Use Azure Language in Foundry Tools to extract PII entities.

☑ Use Azure Language in Foundry Tools to extract named entities.

2 **You want to publish extracts from customer testimonials on a web site. You need to remove personal details from the text before publishing it. What should you do?**

☑ Use Azure Language in Foundry Tools to find and redact PII entities.

☐ Use Azure Language in Foundry Tools to detect the language and publish only the testimonials in English.

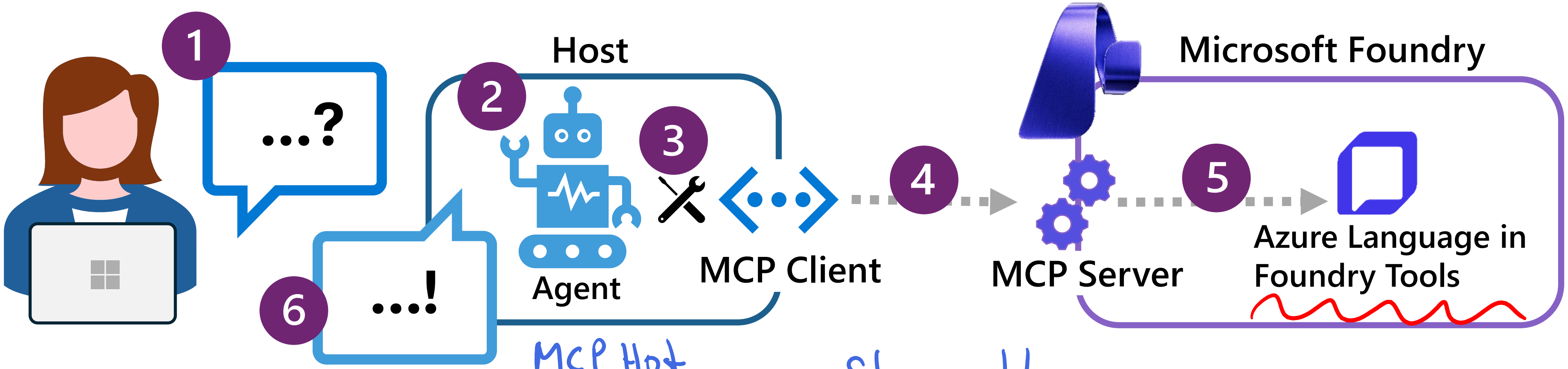
☐ Use a gpt-4.1 model to create new AI-generated customer reviews.

Develop a text analysis agent with the Azure Language MCP server

https://aka.ms/mslearn-language-mcp

Anthropic A2A Google

The Azure Language MCP server



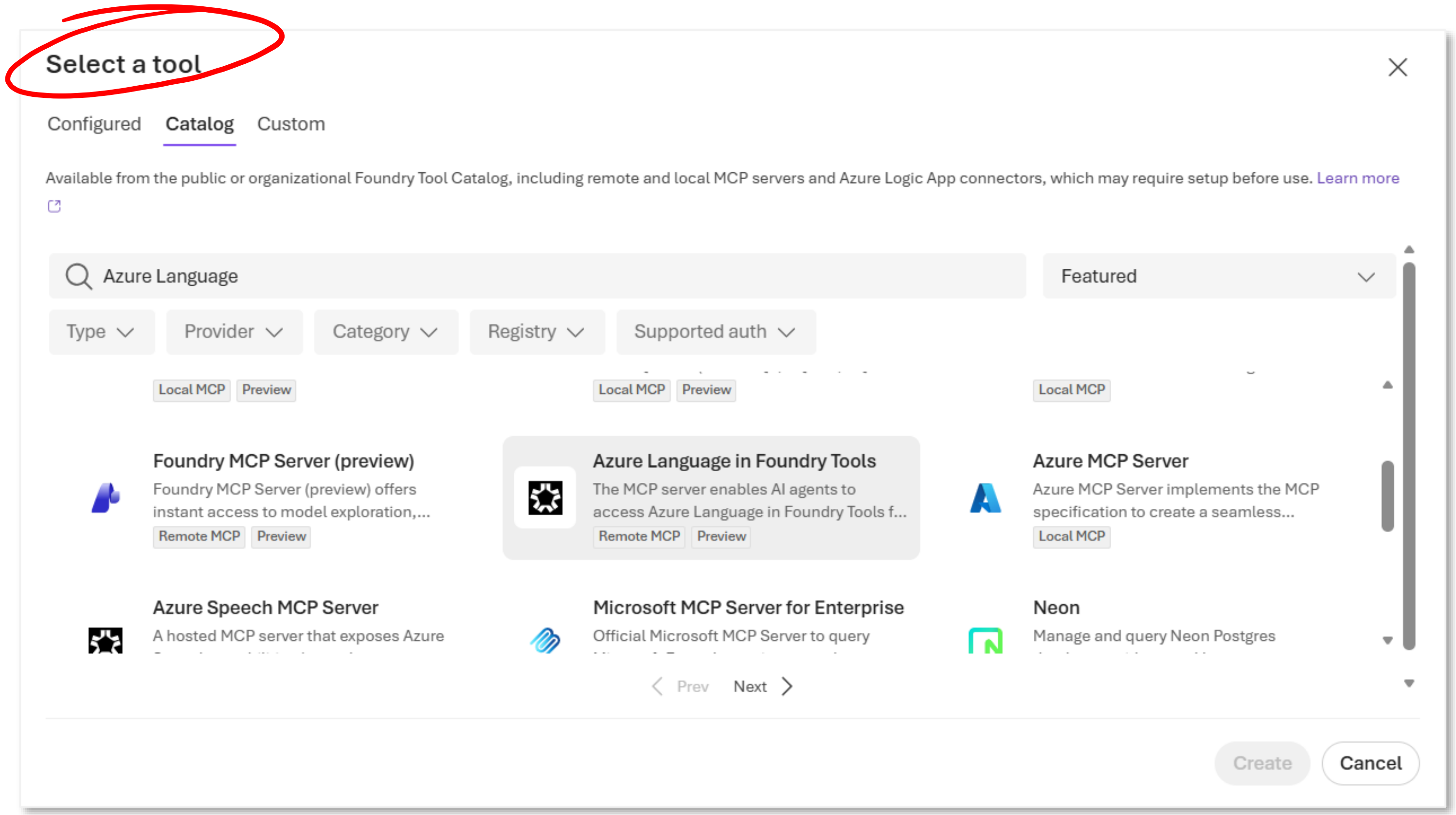
MCP Host

Streamable HTTPS

1. A user sends a prompt to an agent
2. The agent determines task (or tasks) to be performed
3. ~~The agent~~ checks available MCP tools to find the best match (LLM)
4. The agent calls the selected tool through the MCP server, passing the relevant input text
5. The MCP server processes the request using the appropriate Azure Language capability and returns results
6. The agent combines the results into a natural language response for the user

Using the Azure Language MCP server

1. Create an agent
2. Add the Azure Language MCP Server tool
3. Specify agent instructions to select the tool when appropriate
4. Test in the agent playground
5. Build a client application that uses the agent



Exercise

Develop a text analysis agent

In this exercise, you'll:

- Create an agent in Microsoft Foundry
- Add and test the Azure Language MCP tool
- Create a client application for the agent

Start the exercise at:

14. Develop a text analysis agent

1 h

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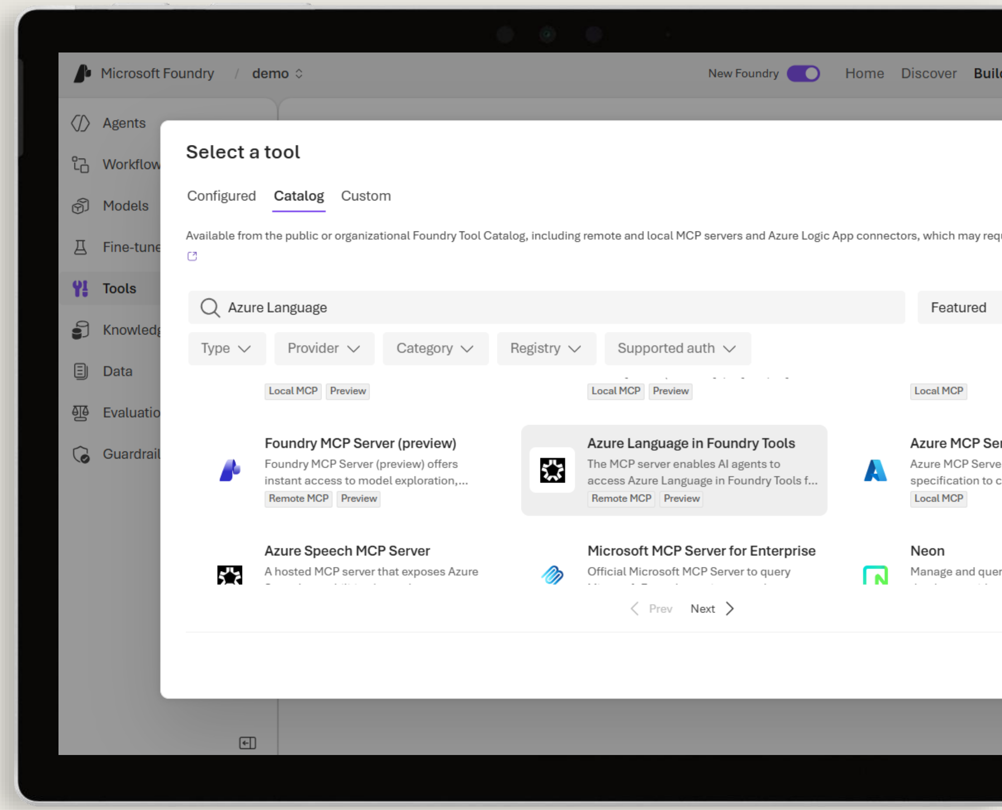


Knowledge check

1 **What is the primary role of the Azure Language MCP server?**

- ☐ To train and fine-tune custom language models for use by AI agents
- ☑ To expose Azure Language text analysis capabilities as MCP tools for agents
- ☐ To deploy and manage large language models in an Azure subscription

2 **How does an agent determine which Azure Language MCP tool to call when processing a user's prompt?**

- ☐ The developer writes routing logic to direct each prompt to a specific tool
- ☑ The agent matches the prompt to tool descriptions received from the MCP server
- ☐ The MCP server analyzes the prompt and automatically routes it to a tool

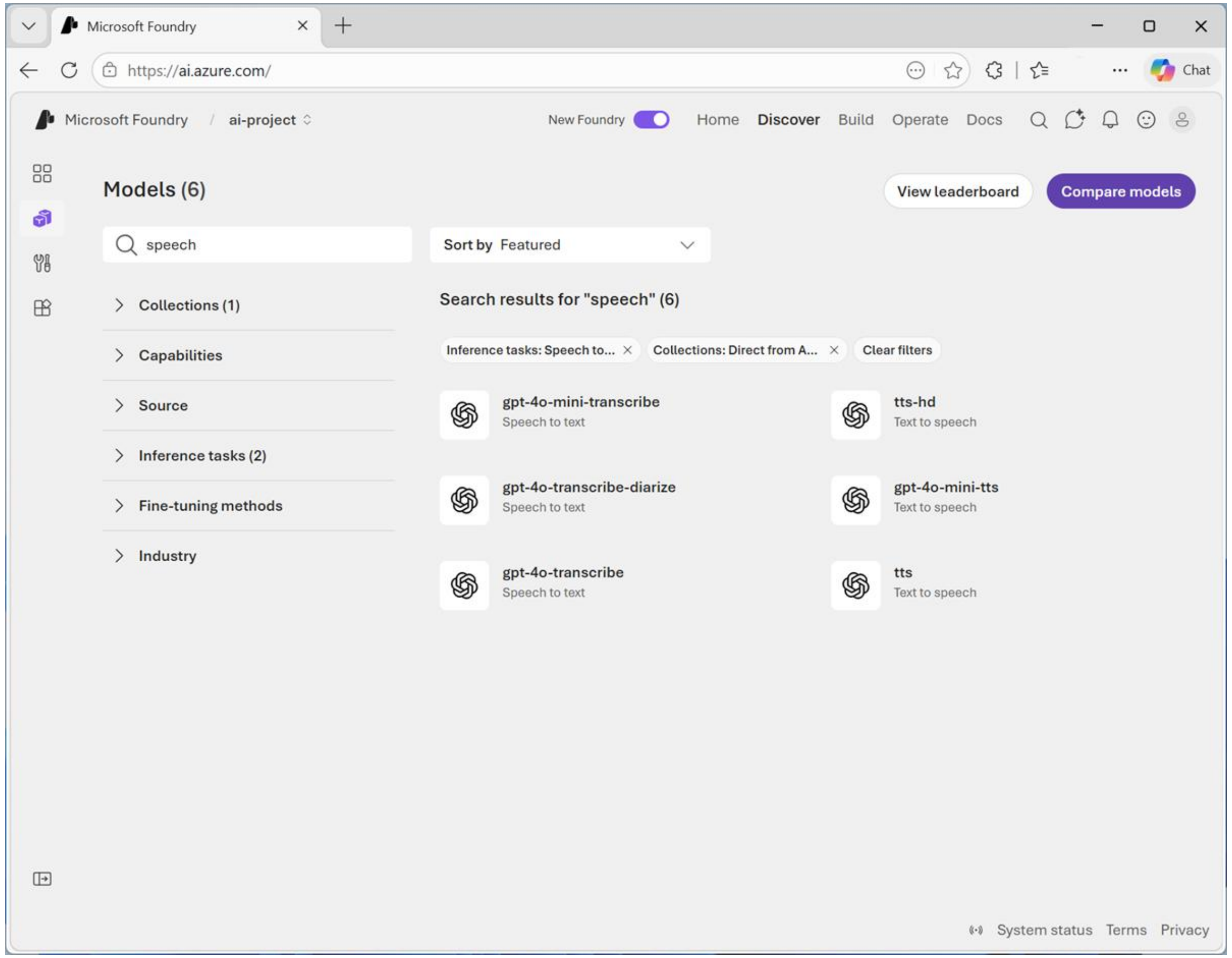
Develop a speech-capable generative AI application

https://aka.ms/mslearn-generative-ai-speech

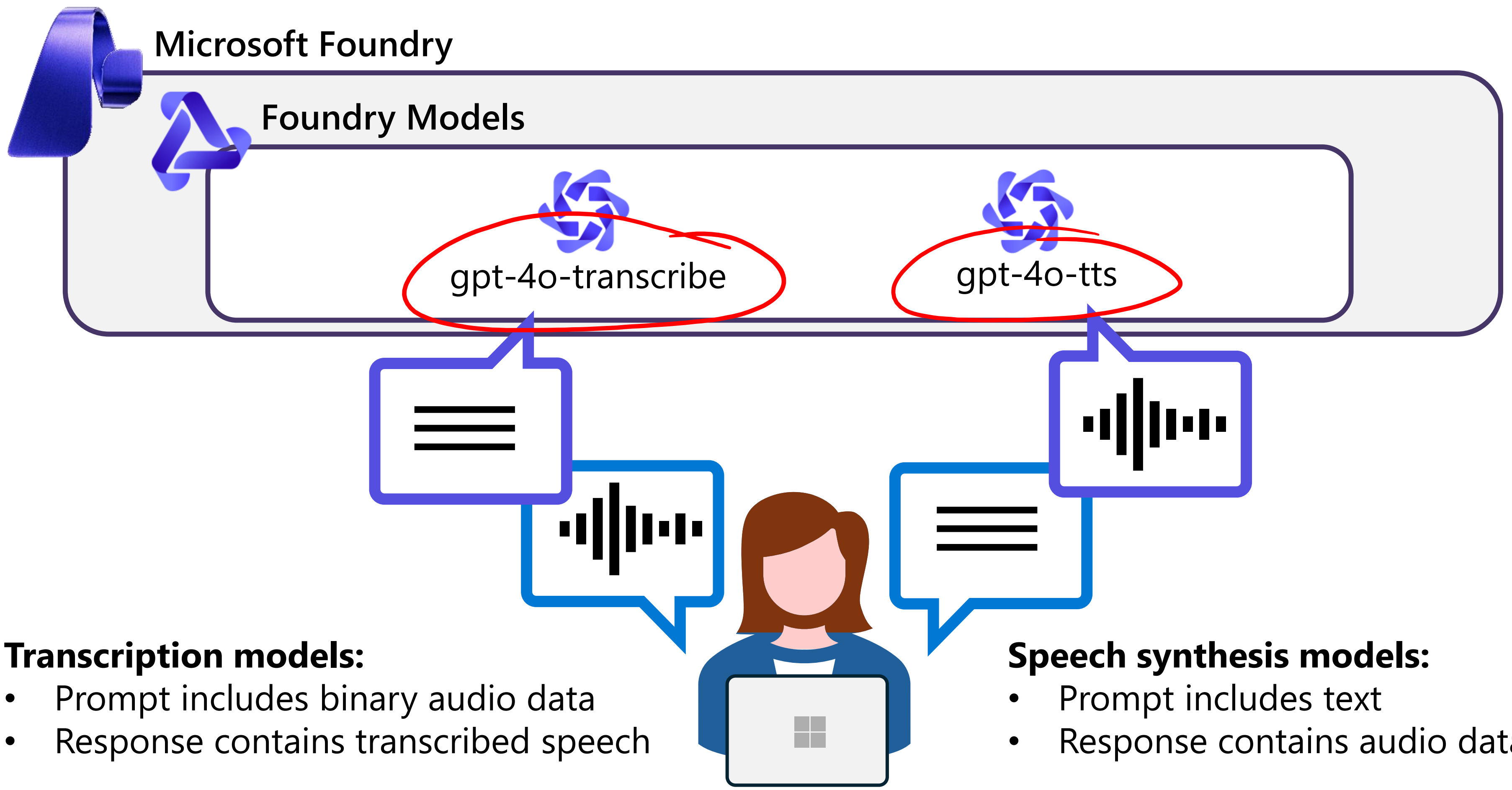
Speech-capable models in Microsoft Foundry Models

Filter and search for:

- Generative AI models that can transcribe speech to text
- Generative AI models that can synthesize text to speech



Using speech-capable generative AI models



Transcription models:
- Prompt includes binary audio data
- Response contains transcribed speech

Speech synthesis models:
- Prompt includes text
- Response contains audio data

Exercise

Use speech-capable generative AI models

In this exercise, you'll:
- Find and deploy speech-capable models in Microsoft Foundry
- Create a speech synthesis app
- Create a speech transcription app

Start the exercise at:

15. Use speech-capable generative AI models

1 h

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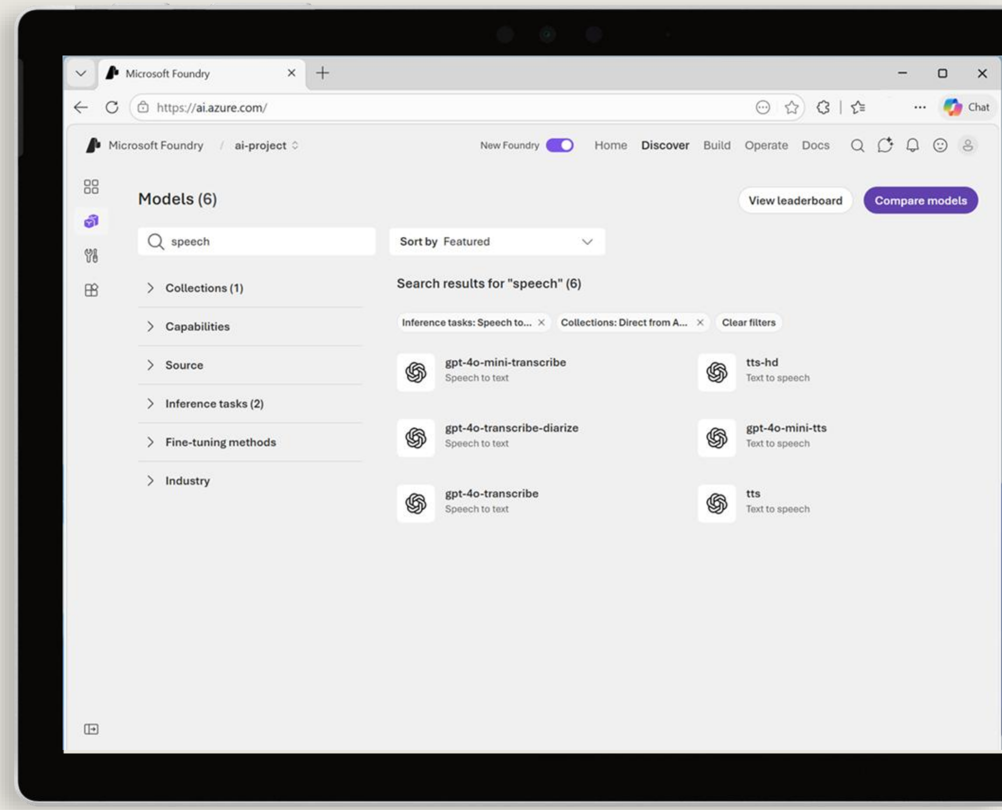


Ollama
LM Studio

Knowledge check

1 **Which model can you use to generate text from speech?**

☐ gpt-4o-mini
☐ gpt-4o-mini-tts
☑ gpt-4o-mini-transcribe

2 **Which model can you use to synthesize speech from text?**

☐ gpt-4o-mini
☑ gpt-4o-mini-tts
☐ gpt-4o-mini-transcribe

Create speech-enabled apps with Microsoft Foundry

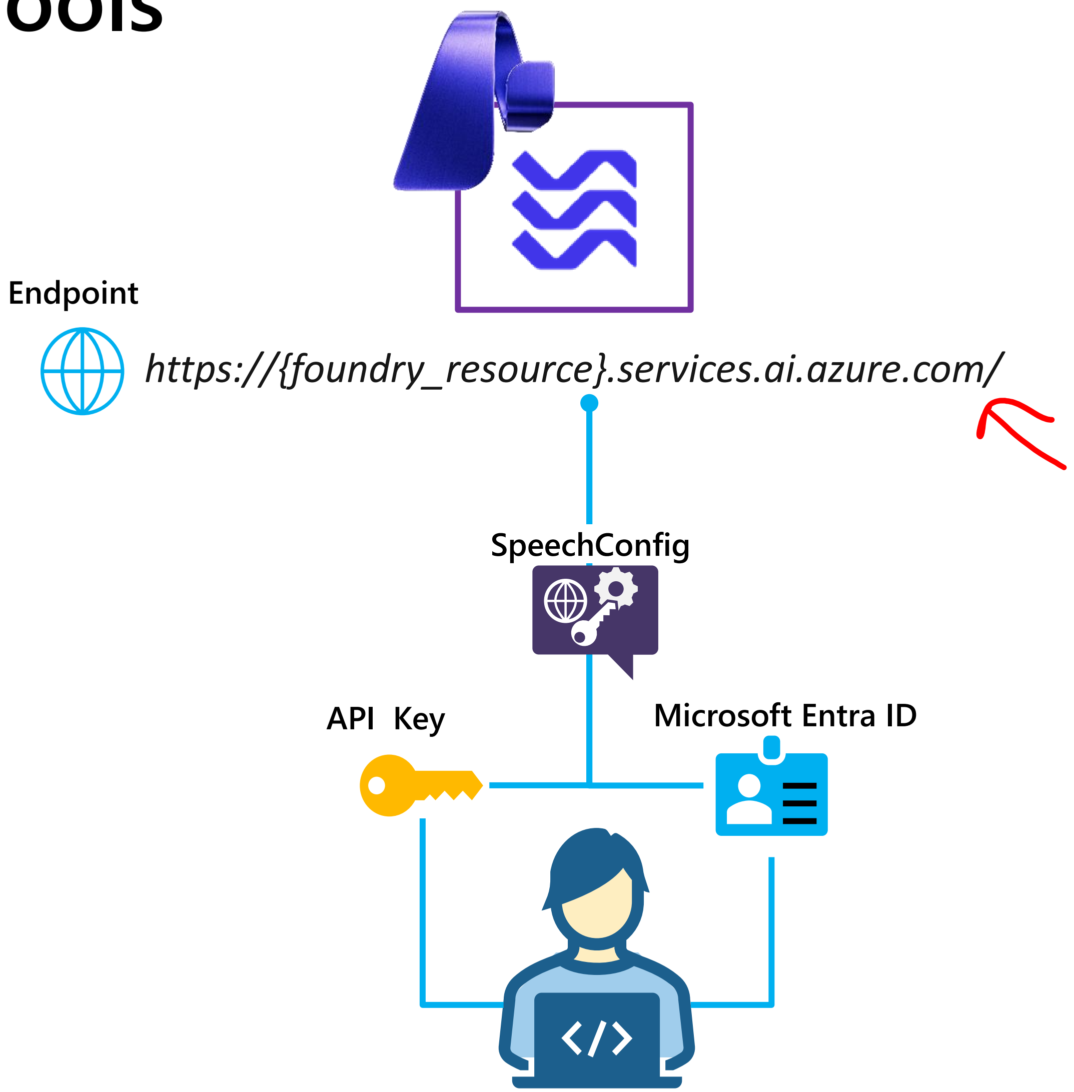
https://aka.ms/mslearn-speech-app

Azure Speech in Foundry Tools

- **Speech to text**: Transcribe spoken audio
- **Text to speech**: Synthesize speech
- **Voice Live**: Real-time conversational speech interactions
- **Speech translation**: Multilingual translation

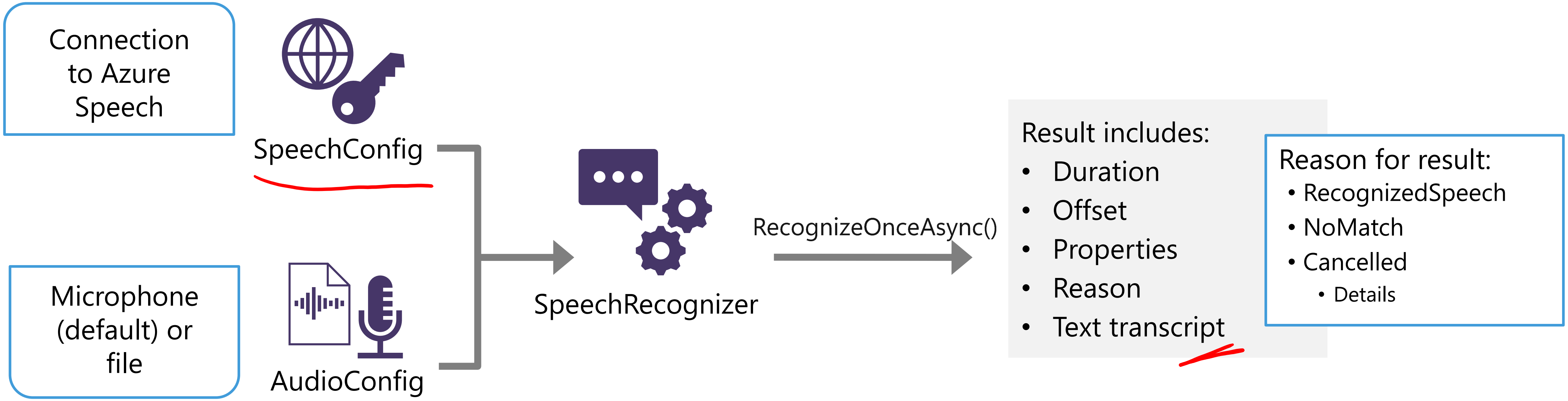
Using Azure Speech in Foundry Tools

1. Connect to the API
   - **Endpoint:**
     - Microsoft Foundry resource*
     - Global/Regional endpoint
   - **Credentials:**
     - Microsoft Foundry project API key
     - Microsoft Entra ID
2. Create a *SpeechConfig* object
   - Proxy object for speech API operations

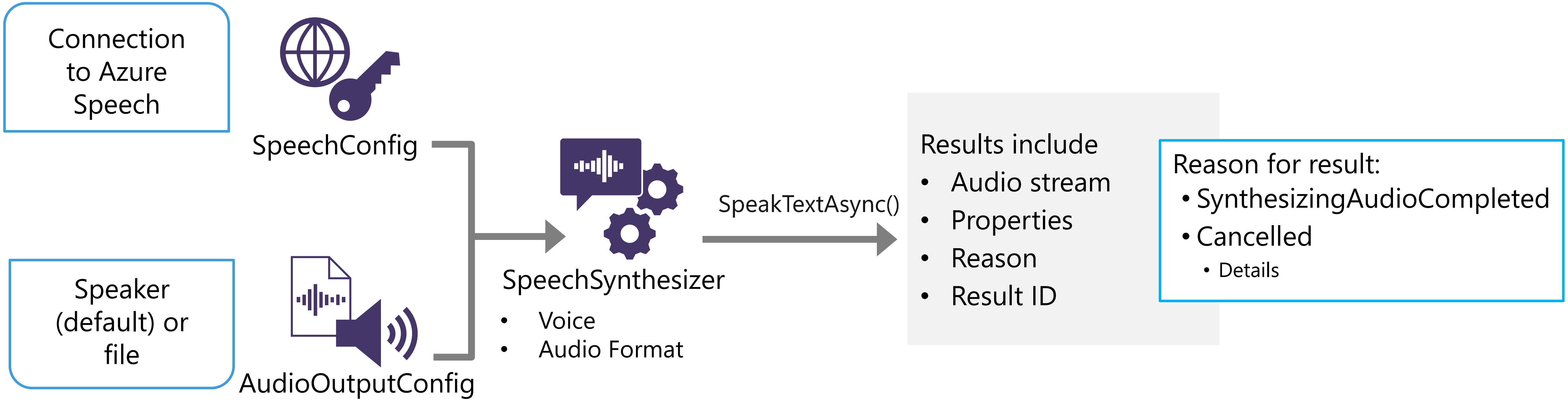


* *cognitiveservices.azure.com is also supported*

Speech to text



Text to speech



Speech Synthesis Markup Language (SSML)

XML-based language with customization options:

- Speaking styles (Neural voices only)
- Pauses and silence
- Phonemes (phonetic pronunciations)
- Prosody (speaking pitch, range, rate, etc.)
- "say-as" (number, date, time, address, etc.)
- Insert recorded speech or background audio

```
SpeakSsmlAsync( ssml-string );
```

```
<speak version="1.0" xmlns="http://www.w3.org/2001/10/synthesis"
                     xmlns:mstts="https://www.w3.org/2001/mstts" xml:lang="en-US">
    <voice name="en-US-AriaNeural">
        <mstts:express-as style="cheerful">
          I say tomato
        </mstts:express-as>
    </voice>
    <voice name="en-US-GuyNeural">
        I say <phoneme alphabet="sapi" ph="t ao m ae t ow"> tomato </phoneme>.
        <break strength="weak"/>Lets call the whole thing off!
    </voice>
</speak>
```

Multiple voices in a single synthesis

Speaking style

Phonetic pronunciation

Pause

Exercise

Recognize and synthesize speech

In this exercise, you'll use Azure Speech to build an app that can:
- Synthesize speech
- Recognize speech

Start the exercise at:

16. Recognize and synthesize speech

No limit

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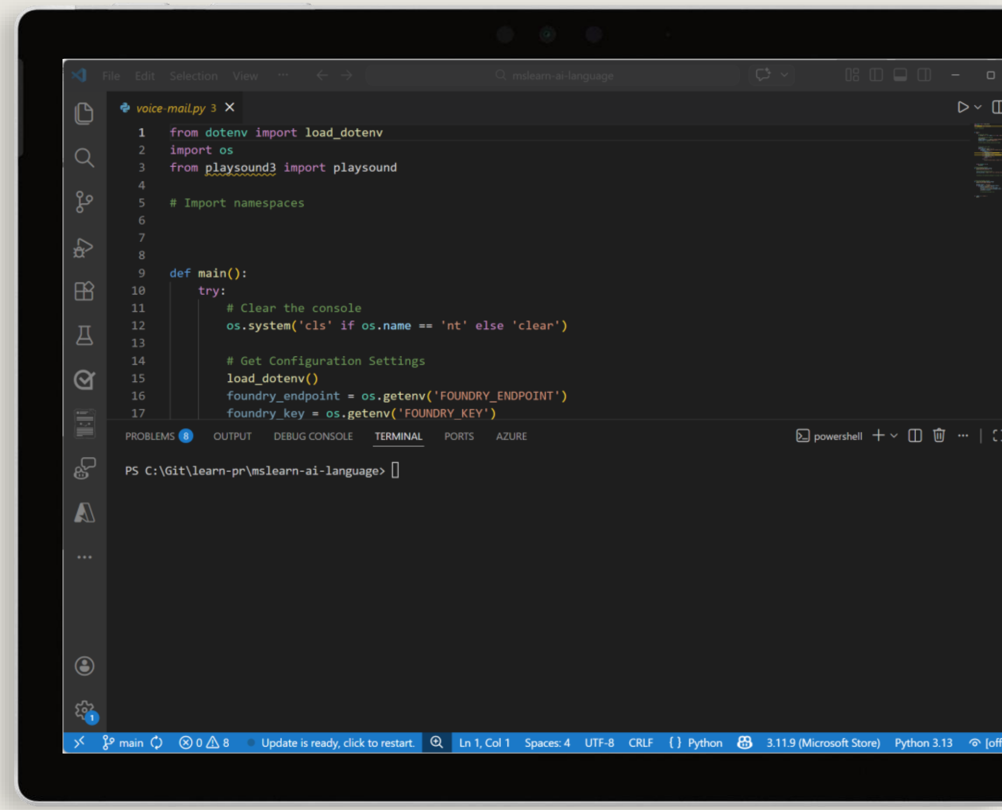


Knowledge check

1 **What information do you need from your Microsoft Foundry resource to consume it using the Azure Speech SDK?**

☑ The endpoint and key
☐ The primary and secondary keys
☐ The Azure subscription ID and resource group name

2 **Which object should you use to specify that the speech input to be transcribed to text is in an audio file?**

☐ SpeechConfig
☑ AudioConfig
☐ SpeechRecognizer

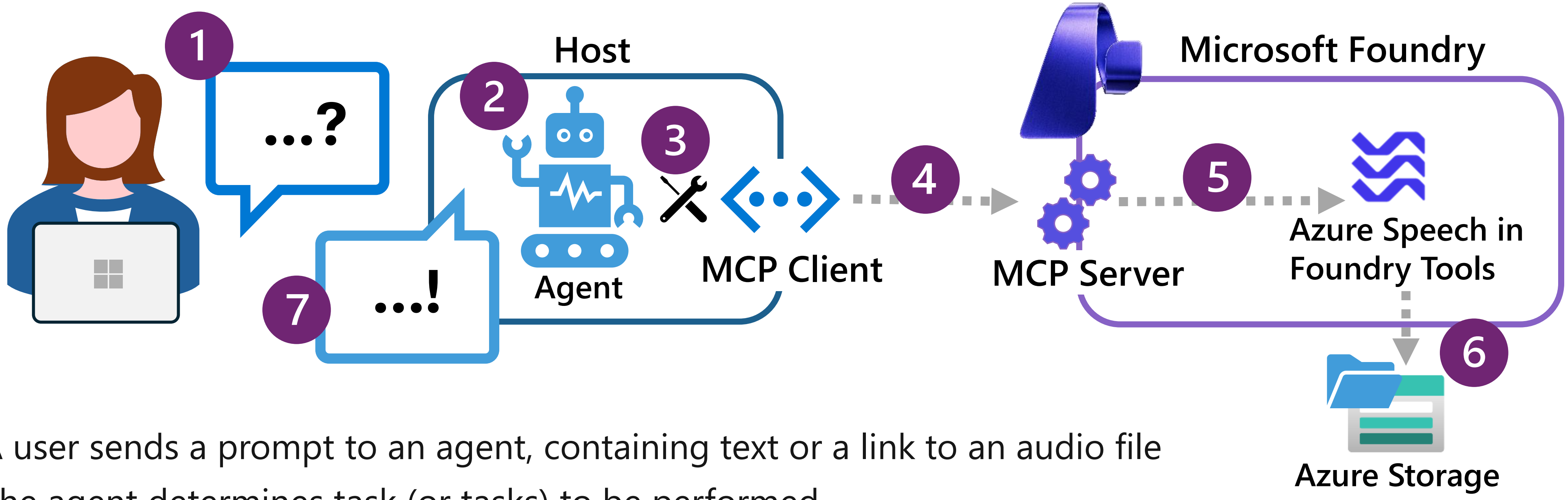
3 **How can you change the voice used in speech synthesis?**

☐ Specify a SpeechSynthesisOutputFormat enumeration in the SpeechConfig object
☑ Set the SpeechSynthesisVoiceName property of the SpeechConfig object to the desired voice name
☐ Specify a filename in the AudioConfig object

Develop a speech agent with the Azure Speech MCP server

https://aka.ms/mslearn-speech-mcp

The Azure Speech MCP server



1. A user sends a prompt to an agent, containing text or a link to an audio file
2. The agent determines task (or tasks) to be performed
3. The agent checks available MCP tools to find the best match

cln 4. The ~~agent~~ calls the selected tool through the MCP server, passing the relevant input

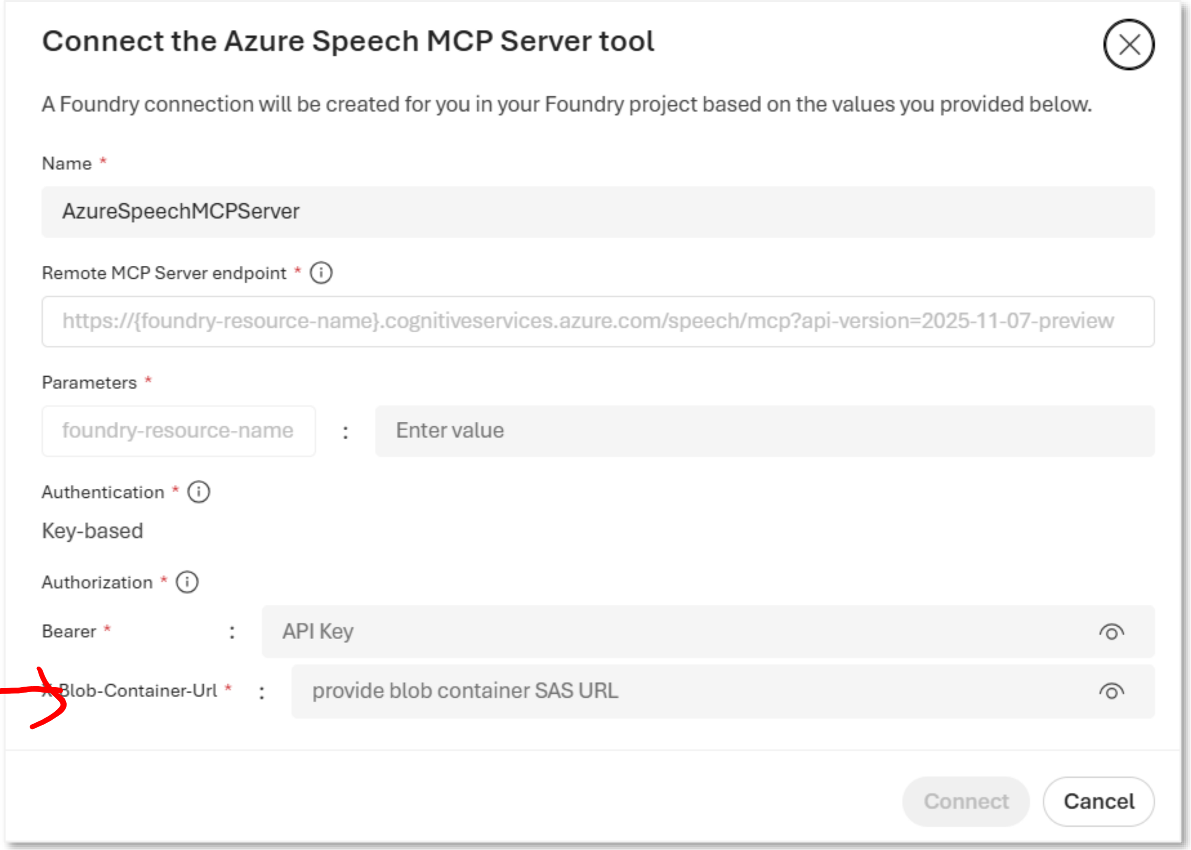
5. The MCP server processes the request using Azure Speech (text to speech or speech to text)
6. In the case of text to speech, the generated audio is saved as a file in Azure storage
7. The agent responds to the user with transcribed text or a link to the generated audio file

Using the Azure Language MCP server

1. Create an agent
2. Connect the Azure Speech MCP Server tool

   Requires an Azure Storage container SAS URL

3. Specify agent instructions to select the tool when appropriate
4. Test in the agent playground
5. Build a client application that uses the agent



Exercise

Use Azure Speech in an agent

In this exercise, you'll:
- Create an agent in Microsoft Foundry
- Add and test the Azure Speech MCP tool
- Create a client application for the agent

Start the exercise at:

17. Use Azure Speech in an agent

1 h

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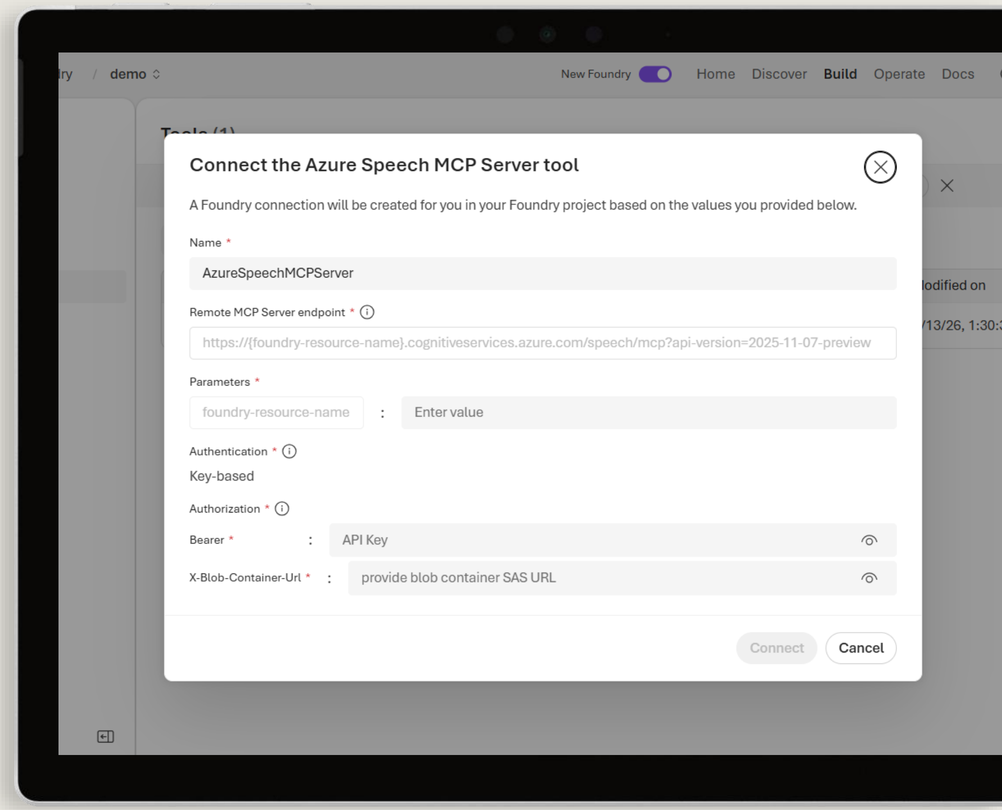


Knowledge check

1 **What two core capabilities does the Azure Speech MCP server expose to agents?**

- ☐ Language translation and text summarization ✗
- ☑ Speech-to-text recognition and text-to-speech synthesis ✓
- ☐ Named entity recognition and sentiment analysis ✗

2 **Why does the Azure Speech MCP server require an Azure Storage account?**

Blob
Files
Queus
Table

HTTPS
SMB

- ☐ To store the agent's instructions and configuration settings ✗
- ☑ To store input audio files and output audio files generated by the speech tools ✓
- ☐ To cache the MCP server's tool definitions for faster discovery ✗

3 **What credentials are needed when connecting the Azure Speech MCP server to a Foundry agent?**

- ☐ An OAuth 2.0 token and a managed identity endpoint URL ✗
- ☑ A Foundry resource key and a SAS URL for a blob container ✓
- ☐ A client certificate and the Azure subscription ID

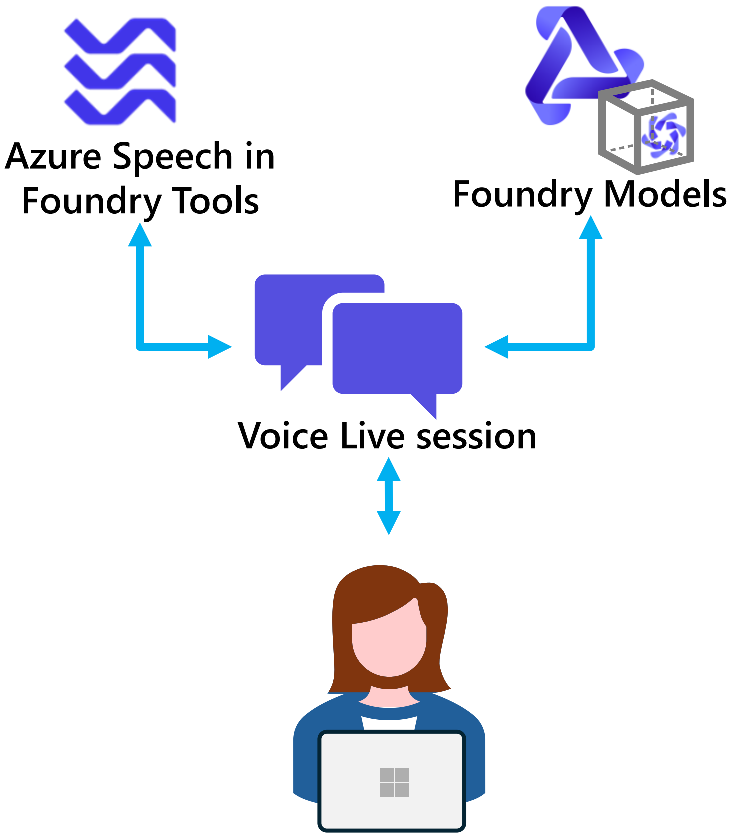
Access Token
von
bis
Audiance https://ai.azure.com

Develop an Azure Speech Voice Live Agent in Microsoft Foundry

https://aka.ms/mslearn-voice-live-agent

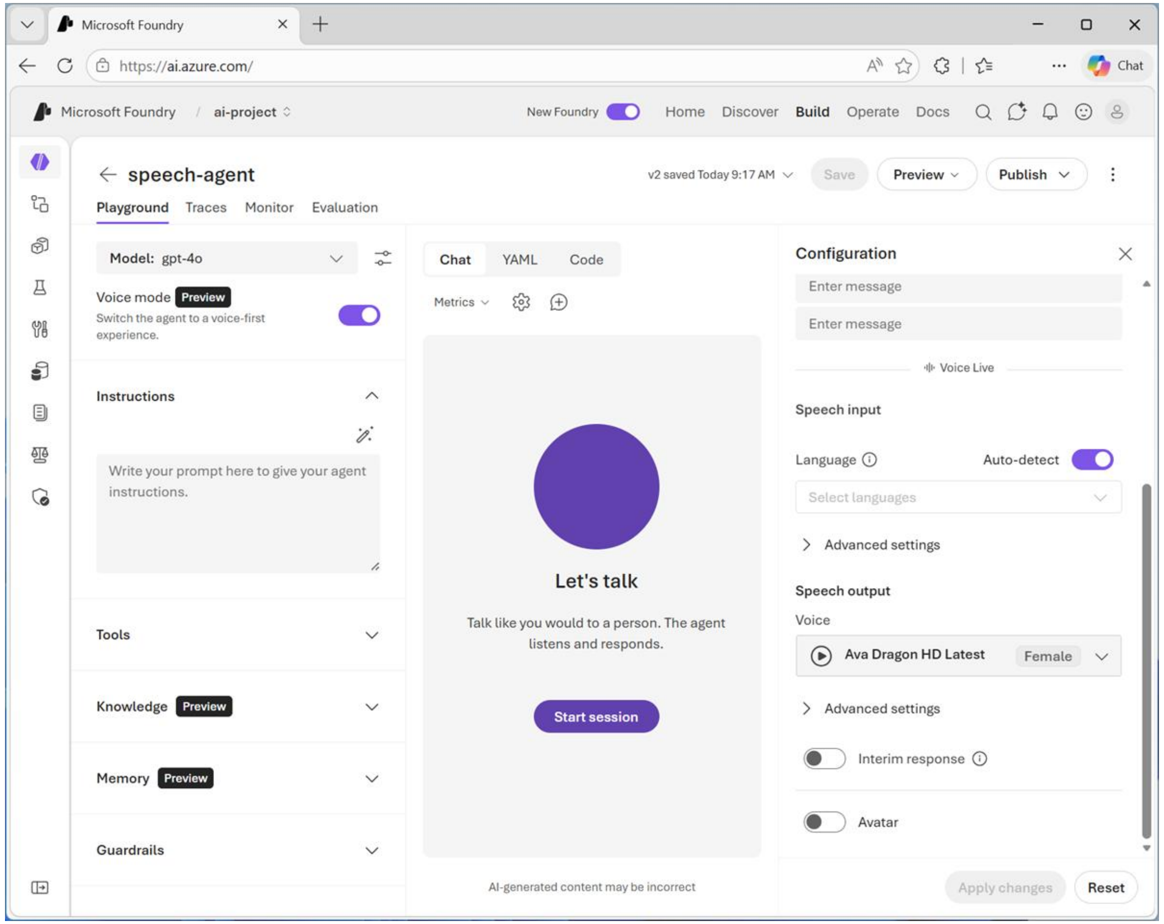
Azure Speech Voice Live

- Real-time, event-based conversation framework
- Connects models to speech conversations in a *Voice Live session*
- Authentication options are:
    - Microsoft Entra ID (keyless)
    - API key
- Advanced features include:
    - Real-time audio processing with support for multiple formats like PCM16 and G.711
    - Voice options, including OpenAI voices and Azure custom voices
    - Avatar integration using WebRTC for video and animation
    - Built-in noise reduction and echo cancellation



Creating a Voice Live agent

- Enable **Audio mode** in the agent playground
  - Configure *Voice Live* settings for the agent
- Create a client app that uses the Voice live SDK to:
  1. Connect to the agent
  2. Configure audio hardware input and output
  3. Establish a *Voice Live* session
  4. Monitor audio systems for activity
  5. Process events (such as user speech input and responses from the agent)



Exercise

Develop a Voice Live agent

In this exercise, you'll:

- Enable and configure Azure Speech Voice Live for an agent
- Implement a conversational app that uses the agent

Start the exercise at:

18. Develop a Voice Live agent

1 h

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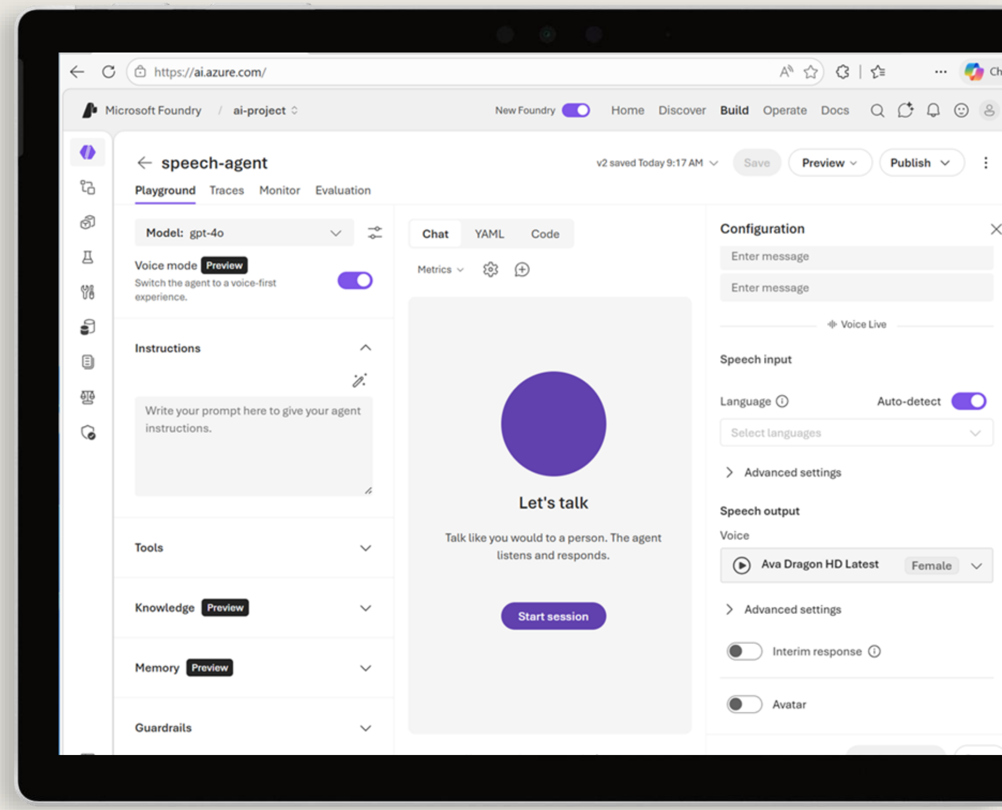


Knowledge check

1 What are the two authentication methods supported by the Voice Live API?

- ☐ OAuth 2.0 and JWT (JSON Web Tokens)
- ☐ Basic authentication and API keys
- ☑ Microsoft Entra (keyless) and API key

2 How do you configure and test Voice Live agent integration in the Foundry Portal?

- ☐ You can't – Voice Live is only accessible through the REST API or Python SDK
- ☐ In the Azure Speech in Foundry Tools Voice Live playground
- ☑ Enable **Voice mode** in the agent playground

3 How can you stop audio playback when a user interrupts the voice agent?

- ☐ You can't – the user must wait for the agent to finish
- ☑ Handle the *INPUT_AUDIO_BUFFER_SPEECH_STARTED* event
- ☐ Reset the Voice Live session and clear the conversation history

Translate text and speech with Microsoft Foundry Tools

https://aka.ms/mslearn-translate

Translation in Microsoft Foundry

Azure Translator in Foundry Tools

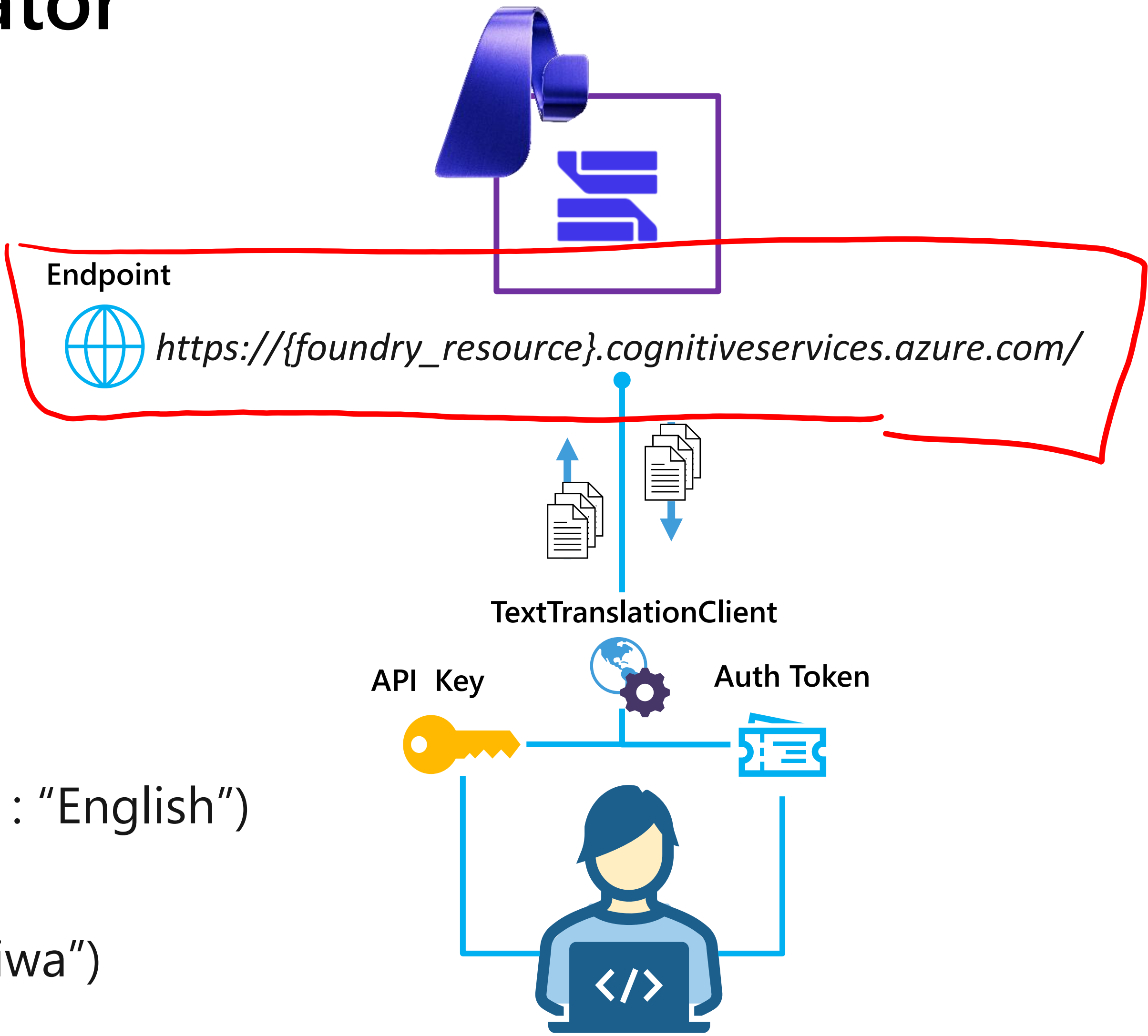
- Multilingual translation for:
  - Text
  - Documents

Azure Speech in Foundry Tools

- Multilingual speech translation:
  - Speech to text
  - Speech to speech

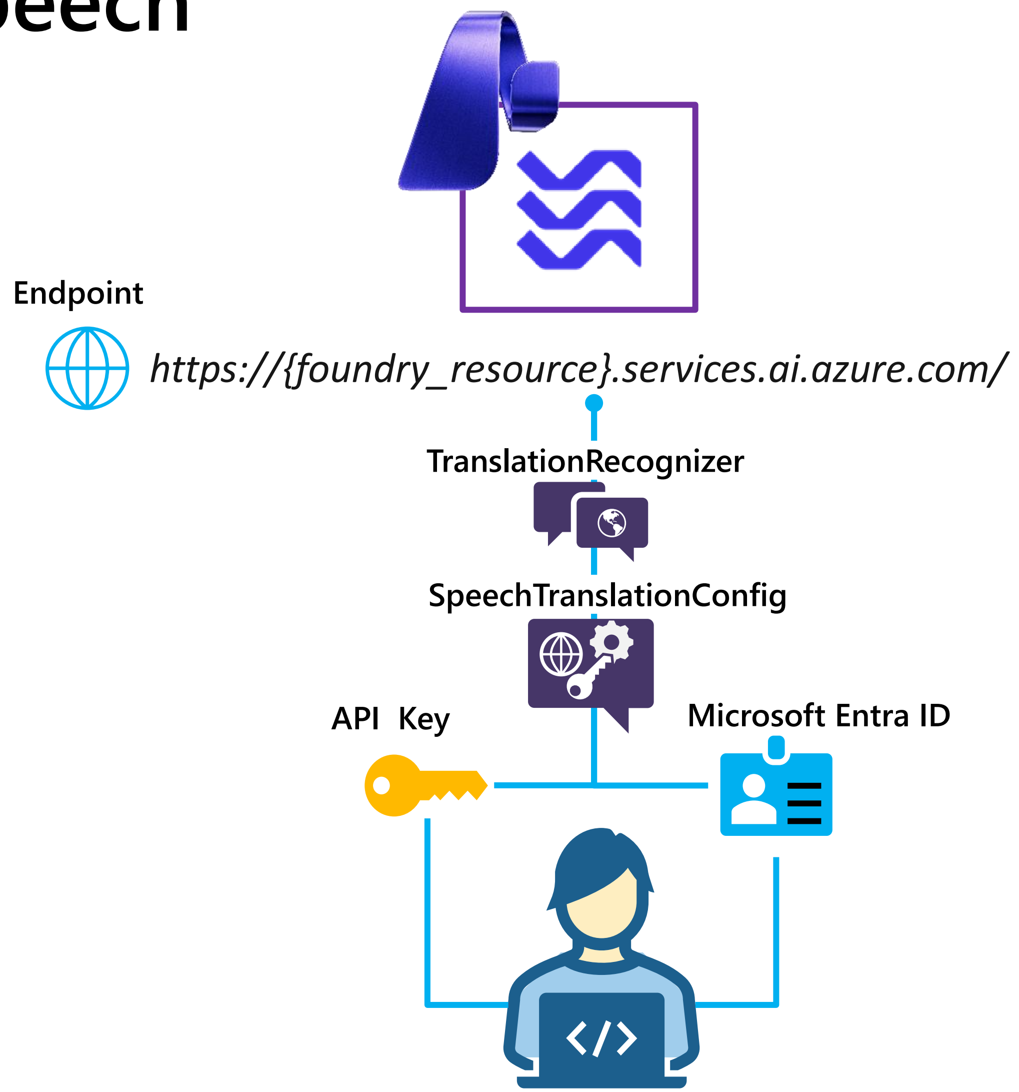
Translate text with Azure Translator

1. Connect to the API
   - **Endpoint:**
     - Microsoft Foundry resource
     - Global / Regional translator endpoint
   - **Credentials:**
     - Microsoft Foundry project API key
     - Authentication token
2. Use a *TextTranslationClient* to perform tasks
   - Get supported languages (for example “en” : “English”)
   - Translate text (“こんにちは” = “Hello”)
   - Transliterate text (“こんにちは” = “Kon'nichiwa”)



Translating speech with Azure Speech

1. Connect to the API
   - **Endpoint:**
     - Microsoft Foundry resource*
     - Global/Regional endpoint
   - **Credentials:**
     - Microsoft Foundry project API key
     - Microsoft Entra ID
2. Create a *SpeechTranslationConfig* object for the connection
3. Use the *TranslationRecognizer* object for translation operations

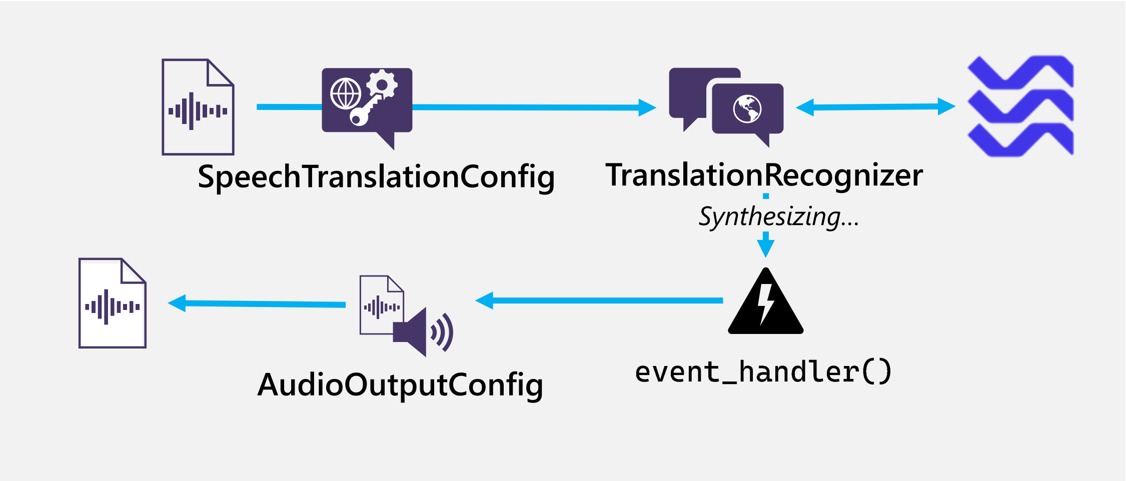


* *cognitiveservices.azure.com is also supported*

Synthesizing Speech Translations

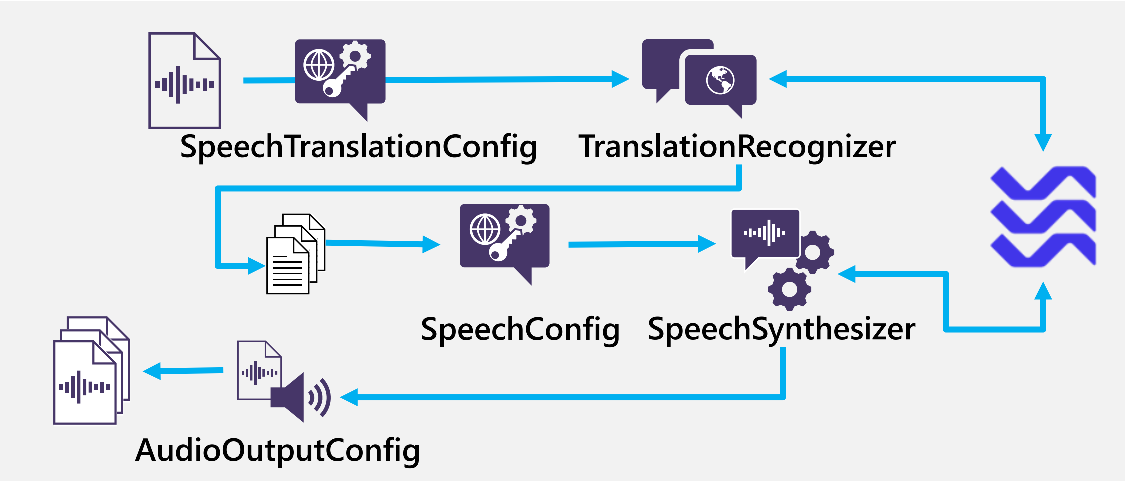
Event-based synthesis

- Only supported for 1:1 translation (single target language)
- Specify desired voice in the **TranslationConfig**
- Use the *Synthesizing* event to retrieve audio stream
- Create an event handler and use *Result.GetAudio()* to retrieve byte stream



Manual synthesis

- Use for multiple target languages
- Translate to text then use Text-to-Speech API to synthesize each translation in the results



Exercise

Translate text and speech

In this exercise, you'll:

- Use Azure Translator in Foundry Tools to translate text
- Use Azure Speech in Foundry Tools to translate speech

Start the exercise at:

19. Translate text and speech

1 h

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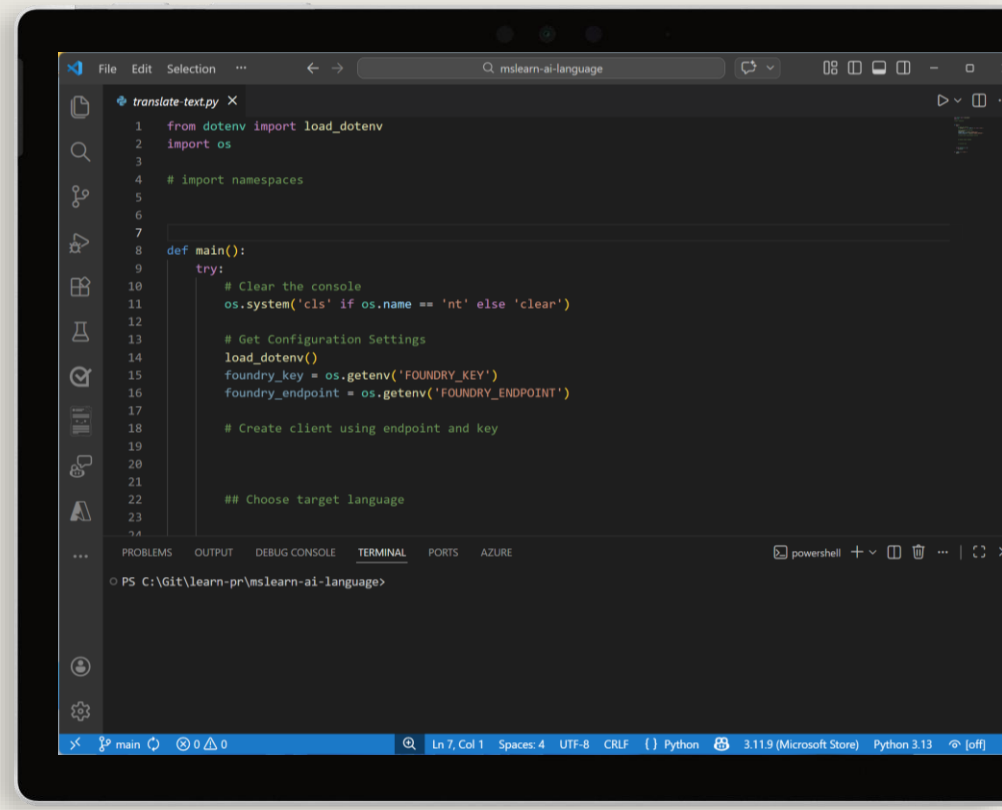

Knowledge check

1. What function of an Azure Translator TextTranslationClient object should you use to convert the Chinese word "你好" to the English word "Hello"?

- [ ] get_supported_language
- [x] translate
- [ ] transliterate

2. What function of an Azure Translator TextTranslationClient object should you use to convert the Russian word "спасибо" in Cyrillic characters to "spasibo" in Latin characters?

- [ ] get_supported_language
- [ ] translate
- [x] transliterate

3. Which Azure Speech SDK object should you use to specify the language(s) into which you want speech translated?

- [ ] SpeechConfig
- [x] SpeechTranslationConfig
- [ ] AudioConfig

Microsoft